

Green's Theorem

Statement · Proof · Verification · Heart-Artery Blockage Case Study · Python Code

Course: Multivariate Calculus **Programme:** Bachelor of Computing & Innovation **Institution:** Riphah International University, Sahiwal

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1. Statement of Green's Theorem

Theorem. Let R be a closed region of the xy -plane bounded by a simple closed curve C . If $M(x, y)$ and $N(x, y)$ are continuous functions of x and y having continuous first partial derivatives in R , then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

This result is called **Green's theorem**. The curve C is traversed **anticlockwise** (positive orientation), so that the region R stays on the left.

Meaning of the symbols

Symbol	Meaning
$\oint_C$	Line integral around the closed contour (boundary curve) C
$\iint_R$	Double integral over the region R enclosed by C (area integral)
M, N	The two components of the vector field $F(x, y) = (M, N) = M \mathbf{i} + N \mathbf{j}$
$\partial N / \partial x - \partial M / \partial y$	The curl (rotation per unit area) of the field in the plane
$dA = dx dy$	Element of area of the region R

In words: the total circulation of the field around the boundary equals the total microscopic rotation (curl) of the field summed over the whole region inside. It links a line integral on the boundary with a double integral over the region.

2. Proof of Green's Theorem

We show that the right-hand side (R.H.S.) of equation (1) equals the left-hand side (L.H.S.) by treating the M-term and the N-term separately. The region is assumed to be describable in two ways: bounded by $y = y_1(x)$ below and $y = y_2(x)$ above for $a \leq x \leq b$ (Fig 1), and by $x = x_1(y)$ on the left and $x = x_2(y)$ on the right for $e \leq y \leq f$ (Fig 2).

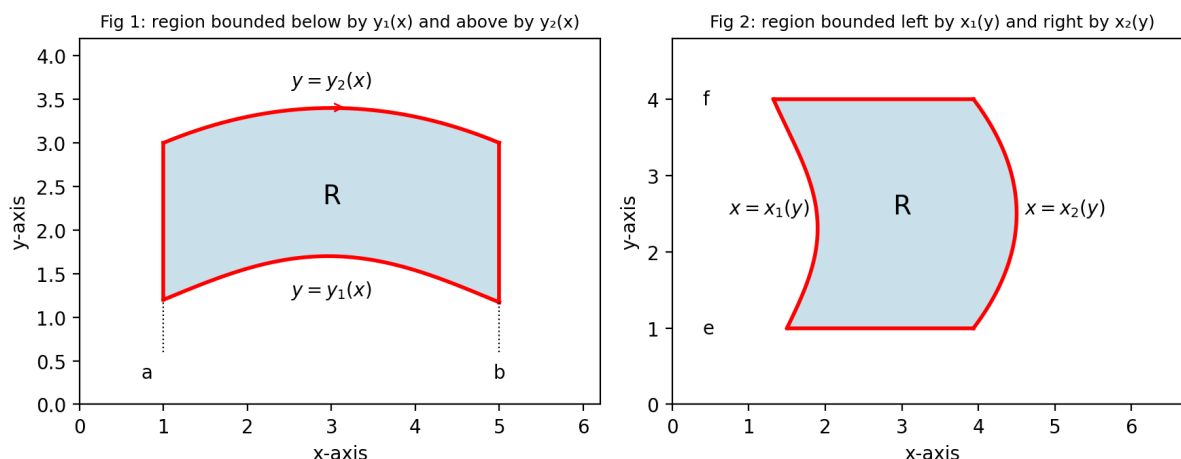


Fig 1 and Fig 2: the two ways of describing the region R

R.H.S. of equation (1):

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Part A - the M-term (using Fig 1)

$$\begin{aligned} \iint_R \frac{\partial M}{\partial y} dx dy &= \int_a^b \left[\int_{y_1(x)}^{y_2(x)} \frac{\partial M}{\partial y} dy \right] dx = \int_a^b [M(x, y)]_{y_1(x)}^{y_2(x)} dx \\ &= \int_a^b [M(x, y_2) - M(x, y_1)] dx = \int_a^b M(x, y_2) dx - \int_a^b M(x, y_1) dx \end{aligned}$$

Reversing the limits of the second term changes its sign, so we write the result as one closed path: along the lower curve y_1 from a to b, then along the upper curve y_2 from b back to a.

$$= - \left[\int_a^b M(x, y_1) dx + \int_b^a M(x, y_2) dx \right] = - \oint_C M dx$$

$$\iint_R \frac{\partial M}{\partial y} dx dy = - \oint_C M dx \quad \Rightarrow \quad - \iint_R \frac{\partial M}{\partial y} dx dy = \oint_C M dx \quad (A)$$

Part B - the N-term (using Fig 2, "similarly")

$$\iint_R \frac{\partial N}{\partial x} dx dy = \int_e^f \left[\int_{x_1(y)}^{x_2(y)} \frac{\partial N}{\partial x} dx \right] dy = \int_e^f [N(x, y)]_{x_1(y)}^{x_2(y)} dy$$

$$= \int_e^f [N(x_2, y) - N(x_1, y)] dy = \int_e^f N(x_2, y) dy + \int_f^e N(x_1, y) dy = \oint_C N dy$$

$$\iint_R \frac{\partial N}{\partial x} dx dy = \oint_C N dy \quad (B)$$

Part C - adding (A) and (B)

$$\oint_C M dx + \oint_C N dy = - \iint_R \frac{\partial M}{\partial y} dx dy + \iint_R \frac{\partial N}{\partial x} dx dy$$

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence proved: L.H.S. = R.H.S. ■

3. Verification of Green's Theorem (worked example)

Problem. Verify Green's theorem for

$$\oint_C (xy + y^2) dx + x^2 dy$$

where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. Here $\mathbf{M} = xy + y^2$ and $\mathbf{N} = x^2$.

Step 0 - intersection points

Solving $x^2 = x$ gives $x(x - 1) = 0$, so the curves meet at **(0, 0)** and **(1, 1)**. Between them $x^2 \leq x$, so the parabola is the lower curve and the line is the upper curve.

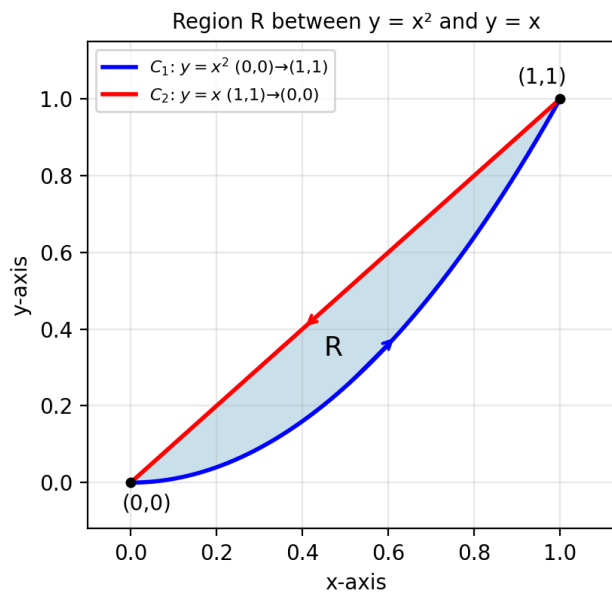


Fig 3: region R and the anticlockwise orientation of C

Step 1 - L.H.S.: line integral along the boundary

(i) Along C_1 : $y = x^2$, from $(0,0)$ to $(1,1)$. Here $dy = 2x \, dx$.

$$\begin{aligned} \int_0^1 [x \cdot x^2 + (x^2)^2] dx + x^2 \cdot 2x \, dx &= \int_0^1 (x^3 + x^4 + 2x^3) dx = \int_0^1 (3x^3 + x^4) dx \\ &= \left[\frac{3x^4}{4} + \frac{x^5}{5} \right]_0^1 = \frac{3}{4} + \frac{1}{5} = \frac{19}{20} \quad (A) \end{aligned}$$

(ii) Along C_2 : $y = x$, from $(1,1)$ back to $(0,0)$. Here $dy = dx$.

$$\int_1^0 (x^2 + x^2) dx + x^2 dx = \int_1^0 3x^2 dx = [x^3]_1^0 = 0 - 1 = -1 \quad (B)$$

(iii) Adding (A) and (B):

$$\oint_C = \frac{19}{20} - 1 = \frac{19 - 20}{20} = -\frac{1}{20} \quad (1)$$

Step 2 - R.H.S.: double integral over the region

With $M = xy + y^2$ and $N = x^2$: $\partial N / \partial x = 2x$ and $\partial M / \partial y = x + 2y$.

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy &= \iint_R (2x - x - 2y) dx \, dy = \iint_R (x - 2y) dx \, dy \\ &= \int_0^1 \int_{x^2}^x (x - 2y) dy \, dx = \int_0^1 [xy - y^2]_{x^2}^x dx = \int_0^1 [(x^2 - x^2) - (x^3 - x^4)] dx \\ &= \int_0^1 (x^4 - x^3) dx = \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = \frac{4 - 5}{20} = -\frac{1}{20} \quad (2) \end{aligned}$$

Conclusion: By (1) and (2), both sides equal $-1/20$. Hence Green's theorem is verified. ✓

4. Case Study: Heart-Artery Blockage Prediction Using Green's Theorem

Assignment #3 – Multivariate Calculus (Max marks 10, time allowed 1.5 hours; instructor: Ghulam Murtaza)

4.1 The problem

Mr. Ahmed, a 52-year-old office worker, has felt **tightness in his chest and shortness of breath** for a few weeks whenever he walks uphill or climbs stairs. At rest he feels normal. Because these are typical warning signs, he visits a cardiac clinic.

4.2 Medical suspicion

After an ECG and basic blood tests the doctors suspect a **partial blockage in one of the coronary arteries**, which may be reducing the oxygen-rich blood reaching the heart muscle. To understand how blood flow is disturbed, they ask a **biomedical engineer** who uses mathematical modelling.

4.3 The engineer's approach

"Blood behaves like a fluid. We can study its behaviour using mathematics. I will model the artery cross-section and analyse how flow changes when there is narrowing." For simplicity:

- The artery is observed as a **2-D circular cross-section**.
- The boundary of the artery is a **closed curve C**.
- The inside region is denoted by **R**.

4.4 Choosing a mathematical model

The engineer chooses a simple vector field to represent blood flow:

$$F(x, y) = (-y, x)$$

- This model represents **rotational (swirling) flow**.
- Such swirling occurs when blood meets resistance due to **plaque build-up**.
- It is a **simplified but effective model** for understanding blockage effects.

4.5 Applying Green's theorem

"Green's theorem tells us that the **blood circulation along the artery wall** is directly related to the **rotation of blood inside the artery**."

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

From the chosen model $M = -y$ and $N = x$, so the curl is:

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1 - (-1) = 2$$

- A **positive curl** means strong (anticlockwise) rotation.
- Strong rotation means blood is **not flowing smoothly**.

Circulation around the wall

Because the curl is the constant 2, Green's theorem gives the circulation around a circular artery wall of radius r directly:

$$\oint_C M dx + N dy = \iint_R 2 dA = 2 \times (\text{area of } R) = 2\pi r^2$$

For the radius $r = 5$ used in the diagrams below, this is $2\pi(25) = 50\pi \approx 157.08$ (in the model's units). A larger value means stronger swirling around the wall.

4.6 Comparing healthy and blocked arteries

Condition	Flow behaviour	Curl
Healthy artery	Smooth flow, almost no rotation	$= 0$
Partially blocked artery	Swirling flow, high rotation	$\neq 0$ (large)

Mr. Ahmed's modelled flow shows **high rotation** (curl = 2), matching the blocked-artery condition.

4.7 Medical interpretation

“Because circulation around the artery is disturbed and internal rotation is high, this suggests a **partial blockage**. Blood is struggling to pass smoothly, reducing oxygen supply to the heart.” This mathematical insight supports the medical imaging results.

4.8 Final outcome

Doctors use this information to:

- Confirm the seriousness of the blockage
- Recommend further imaging
- Plan treatment such as **medication or stent placement**

Task: Write a Python code for this problem using if-else conditions and also draw the diagram. (Solved in Section 5.)

Important note: this case study is a *teaching model*. Real arterial blood flow is 3-D, pulsatile and governed by the Navier–Stokes equations; clinicians diagnose blockages with imaging and tests such as angiography, not with a single curl value. The exercise shows how *circulation and curl* can be interpreted physically.

5. Python Implementation (if / elif / else + diagrams)

The three cases from the lecture are combined into one reusable function `analyze(M, N)`. Changing the vector field changes the output and the diagram automatically – this is exactly the **Home Task** (“if the function input changes, the output changes”).

Decision rule

Case	Field $F(x, y)$	Curl	Condition	Interpretation
1	$(-y, x)$	2	$\text{curl} > 0$	Anticlockwise rotation – possible artery blockage
2	(x, y)	0	$\text{curl} = 0$	Smooth blood flow – healthy artery
3	$(y, -x)$	-2	$\text{curl} < 0$	Clockwise rotation – abnormal blood flow

Complete program

```
import numpy as np
import matplotlib.pyplot as plt
from sympy import symbols, diff

x, y = symbols('x y')

def analyze(M, N, label, save_as=None):
    """Green's theorem curl test for  $F(x, y) = (M, N)$ ."""
    # Curl formula:  $dN/dx - dM/dy$ 
    dN_dx = diff(N, x)
    dM_dy = diff(M, y)
    curl = dN_dx - dM_dy

    # Decision using if-elif-else
    if curl > 0:
        condition = "Positive Curl"
        interpretation = "Anticlockwise rotation, possible artery blockage."
        colour, title = "red", "Curl > 0 : Anti-Clockwise Rotation (Blocked Artery)"
    elif curl == 0:
        condition = "Zero Curl"
        interpretation = "Smooth blood flow (Healthy artery)."
        colour, title = "green", "Curl = 0 : Smooth Blood Flow (Healthy Artery)"
    else:
        condition = "Negative Curl"
        interpretation = "Clockwise rotation, abnormal blood flow."
        colour, title = "red", "Curl < 0 : Clockwise Rotation (Abnormal Flow)"

    # Output
    print(f"--- {label} ---")
    print(f"Vector Field:  $F(x, y) = (M, N)$ ")
    print(f"dN/dx =", dN_dx)
    print(f"dM/dy =", dM_dy)
    print(f"Curl =", curl)
    print(f"Condition:", condition)
    print(f"Medical Interpretation:", interpretation)
    print()

    # Diagram: vector field + artery boundary (circle of radius 5)
    xv, yv = np.meshgrid(np.linspace(-6, 6, 20), np.linspace(-6, 6, 20))
    Mv = np.broadcast_to(float(M) if M.is_number else
                          np.vectorize(lambda a, b: float(M.subs({x: a, y: b}))) (xv, yv),
                          xv.shape)
    Nv = np.broadcast_to(float(N) if N.is_number else
                          np.vectorize(lambda a, b: float(N.subs({x: a, y: b}))) (xv, yv),
                          xv.shape)
```

```

theta = np.linspace(0, 2 * np.pi, 200)
radius = 5

plt.figure(figsize=(6, 6))
plt.quiver(xv, yv, Mv, Nv)
plt.plot(radius * np.cos(theta), radius * np.sin(theta), colour, linewidth=2)
plt.title(title)
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.axis("equal")
plt.grid(True)
if save_as:
    plt.savefig(save_as, dpi=150, bbox_inches="tight")
plt.close()
return curl

# Case 1: F = (-y, x) -> curl = 2 (blocked artery)
analyze(-y, x, "Case 1", "case1.png")
# Case 2: F = (x, y) -> curl = 0 (healthy artery)
analyze(x, y, "Case 2", "case2.png")
# Case 3: F = (y, -x) -> curl = -2 (abnormal flow)
analyze(y, -x, "Case 3", "case3.png")

```

Program output

```

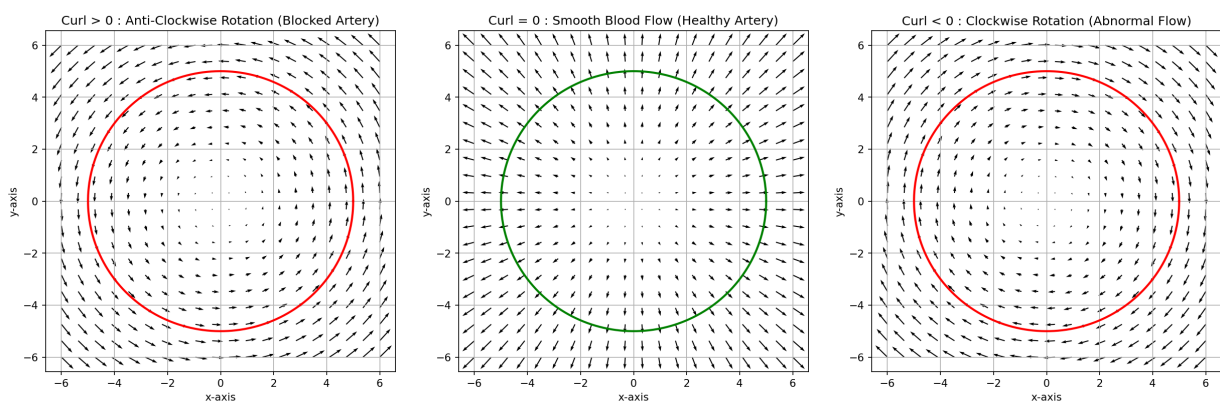
--- Case 1 ---
Vector Field: F(x, y) = (-y, x)
dN/dx = 1
dM/dy = -1
Curl = 2
Condition: Positive Curl
Medical Interpretation: Anticlockwise rotation, possible artery blockage.

--- Case 2 ---
Vector Field: F(x, y) = (x, y)
dN/dx = 0
dM/dy = 0
Curl = 0
Condition: Zero Curl
Medical Interpretation: Smooth blood flow (Healthy artery).

--- Case 3 ---
Vector Field: F(x, y) = (y, -x)
dN/dx = -1
dM/dy = 1
Curl = -2
Condition: Negative Curl
Medical Interpretation: Clockwise rotation, abnormal blood flow.

```

Diagrams produced



Case 1: $\text{curl} = 2 > 0$ – anticlockwise swirl (possible blockage)

Case 2: $\text{curl} = 0$ – outward flow, no rotation (healthy)

Case 3: $\text{curl} = -2 < 0$ – clockwise swirl (abnormal flow)

In each plot the black arrows are the vector field $F = (M, N)$ and the coloured circle (radius 5) is the artery boundary C .

6. Formulas: Gradient, Divergence and Curl

Using the del operator $\nabla = \partial/\partial x \mathbf{i} + \partial/\partial y \mathbf{j}$ and a field $F(x, y) = (M, N)$:

Operator	Notation	Formula (2-D)	What it tells us
Gradient	∇f	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$	Direction and rate of fastest increase of a scalar function – used for speed / change
Divergence	$\nabla \cdot F$	$\nabla \cdot F = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$	How much the field spreads out of (source) or into (sink) a point
Curl	$\nabla \times F$	$(\nabla \times F)_z = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$	Rotation (clockwise / anticlockwise) – used to check blockage

Worked example from the lecture: $F(x, y) = (-x, -y)$

Here $F = -x \mathbf{i} + (-y) \mathbf{j}$, so $M = -x$ and $N = -y$.

$$\nabla \cdot F = \frac{\partial(-x)}{\partial x} + \frac{\partial(-y)}{\partial y} = -1 - 1 = -2, \quad \nabla \times F = \frac{\partial(-y)}{\partial x} - \frac{\partial(-x)}{\partial y} = 0 - 0 = 0$$

This field points straight toward the origin (a sink): it has divergence but **no rotation**, so its curl is zero. Remember that ∇ defines the **direction** and not only a numeric value.

Using these quantities for blood flow

- **Gradient** – used for the speed (rate of change) of flow.
- **Curl** – used to calculate rotation (clockwise / anticlockwise).
- If there is a blockage, there will be rotation; if the curl value is **positive**, a blockage is indicated.
- **Curl ≤ 0** (in the lecture's model) $\rightarrow$ no blockage indicated.

Caveat: in the programme above, a negative curl is reported as “abnormal rotation” (clockwise), whereas the lecture summary treats curl ≤ 0 as “no blockage”. Both follow from the same value; the difference is only in the interpretation rule the instructor chooses, so state your rule clearly in your answer.

7. Notes, Real-Life Use and Home Task

Key notes from the lecture

- Green's theorem is used to find out blood blockage in arteries (as modelled in the case study).
- The boundary (closed-curve) line integral of the field equals the double integral of the curl over the region enclosed by that boundary.
- In particular, choosing $M = -y/2$ and $N = x/2$ gives curl = 1, so the line integral **equals the area of R**.

Question 1 – How is Green's theorem used in real life?

- **Area computation:** surveyors' planimeters and polygon-area algorithms in GIS / CAD / computer graphics use the boundary form $A = \frac{1}{2} \oint (x dy - y dx)$.

- **Fluid flow & aerodynamics:** relating circulation around a wing or pipe wall to vorticity inside.
- **Physics & engineering:** work done by a force around a closed path; electromagnetism (2-D form of Stokes' theorem).
- **Biomedical modelling:** interpreting swirling flow (curl) in a vessel cross-section, as in the artery case study.

Question 2 - What is the real-life purpose of Green's theorem?

Its purpose is to **convert a hard integral into an easier one** – a difficult line integral around a complicated closed curve can be replaced by a double integral over the region (or vice versa) – and to give a physical link between **boundary circulation** and **internal rotation**.

Home task (from the lecture)

- Make a Python code so that if the function is changed, the output changes accordingly (Section 5, function analyze(M, N)).
- Draw the diagram with Python (matplotlib quiver plot with the artery boundary circle).

Summary

1. Green's theorem: $\oint (M dx + N dy) = \iint (\partial N / \partial x - \partial M / \partial y) dx dy$.
2. Proof: split into the M-part ($-\iint \partial M / \partial y = \oint M dx$) and the N-part ($\iint \partial N / \partial x = \oint N dy$), then add.
3. Verified for $M = xy + y^2$, $N = x^2$ over the region between $y = x$ and $y = x^2$: both sides = $-1/20$.
4. Case study: $F = (-y, x)$ gives $\text{curl } 2 > 0 \rightarrow$ anticlockwise swirl $\rightarrow$ possible blockage; $F = (x, y)$ gives $\text{curl } 0 \rightarrow$ smooth flow; $F = (y, -x)$ gives $\text{curl } -2 \rightarrow$ clockwise abnormal flow.
5. Python: sympy computes the curl, if-elif-else decides the condition, matplotlib draws the diagram.